Practical 3

Aim: Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge toclassify a new sample

Source-Code:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Sample Dataset
data = {
   'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain',
            'Overcast','Sunny','Sunny','Rain','Sunny',
            'Overcast','Overcast','Rain'],

   'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool',
               'Cool','Mild','Cool','Mild','Mild',
               'Mild','Hot','Mild'],

   'Humidity':['High','High','High','High','Normal','Normal',
            'Normal','High','Normal','Normal','Normal',
            'High','Normal','High'],

   'Wind':['Weak','Strong','Weak','Weak','Weak','Strong',
         'Strong','Weak','Weak','Weak','Strong',
         'Strong','Weak','Strong'],

   'Play':['No','No','Yes','Yes','Yes','No',
         'Yes','No','Yes','Yes','Yes',
         'Yes','Yes','No']
}

df = pd.DataFrame(data)

# Label Encoding
encoder = LabelEncoder()

for column in df.columns:
   df[column] = encoder.fit_transform(df[column])

X = df.drop('Play', axis=1)
y = df['Play']

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, random_state=42)

# Decision Tree using ID3 (Entropy)
model = DecisionTreeClassifier(criterion='entropy', random_state=42)
```

```
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

# Display Decision Tree
print("\nDecision Tree Rules:\n")
print(export_text(model, feature_names=list(X.columns)))

# Predict New Sample
new_sample = [[2, 2, 1, 1]]

prediction = model.predict(new_sample)

if prediction[0] == 1:
   print("\nPrediction: Play = Yes")
else:
   print("\nPrediction: Play = No")
```

Output:

```
Accuracy: 1.0

Decision Tree Rules:

|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook >  0.50
|   |--- Humidity <= 0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- class: 0
|   |   |--- Wind >  0.50
|   |   |   |--- Outlook <= 1.50
|   |   |   |   |--- class: 1
|   |   |   |--- Outlook >  1.50
|   |   |   |   |--- class: 0
|   |--- Humidity >  0.50
|   |   |--- Wind <= 0.50
|   |   |   |--- Outlook <= 1.50
|   |   |   |   |--- class: 0
|   |   |   |--- Outlook >  1.50
|   |   |   |   |--- class: 1
|   |   |--- Wind >  0.50
|   |   |   |--- class: 1


Prediction: Play = Yes
```